Advanced Macroeconomics

I. Foundations of Dynamic Macroeconomic Modeling
   1. Introduction
   2. History of Macroeconomics
   3. Static General Equilibrium Models
   4. Saving and Investment in a Two-Period Model

II. Long-run Economic Growth

III. Short-run Fluctuations

IV. Applications

Can Pessimism Cause a Recession?

Households become worried about their economic future.

⇓

They reduce consumption and increase desired saving.

⇓

What happens to GDP and employment?

- Will the economy return to its previous output level without policy intervention?
- Which economic mechanisms determine your answer?

2. History of Macroeconomics

Traditional Macroeconomics
Modern Macroeconomics
Criticisms of Macroeconomics
Summary and Literature

The Time Before Macroeconomics

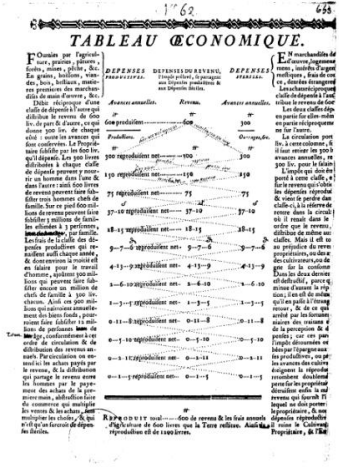
- François QUESNAY (1694–1774): Tableau Economique, economy as circular flow
- Pre-KEYNESian period: partial equilibrium and aggregate analysis not distinguished (Adam SMITH, Karl MARX, Alfred MARSHALL)
- Beginning of the Neoclassical Era around 1870: marginal analysis, individual optimization, and the development of general equilibrium theory
- Léon WALRAS (1834-1910): General Equilibrium Theory, Formalization of the „invisible hand“, competitive general equilibrium under idealized conditions
- Knut WICKSELL (1851-1926): Reformulation of the Quantity Theory (endogenous money stock, endogenous velocity of money)
- Irving FISHER (1867-1947): The Theory of Interest

Francois Quesnay: Tableau Economique

Francois QUESNAY (1694-1774)



(Wikipedia)



(1759)

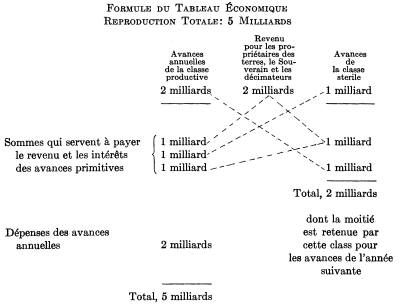
Tableau Economique (1766)

(Gilbert 1989)

- An early model of an interdependent economic system
- The system of production and consumption is conceived as a circular process
  - The same goods can appear both among the products and among the means of production
  - Human consumption itself, apparently the final purpose of any productive activity, is determined by the previous production
- When the economy produces more than the minimum necessary to restore depleted reserves, a surplus results: Net product

Tableau Economique and Input-Output-Analysis

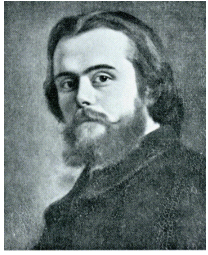
(Phillips 1955)



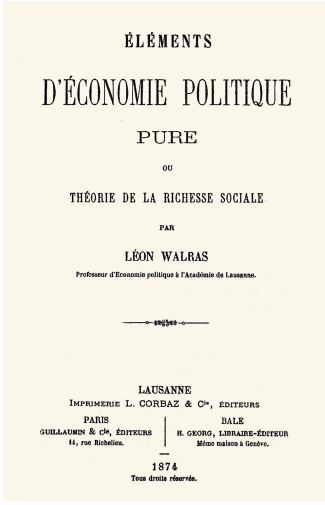
	Purchasing Industry			Total
Producing Industry	Farmers	Artisans	Proprietors	Production
Farmers	2	2	1	5
Artisans	1	0	1	2
Proprietors	2	0	0	2
Total Production	5	2	2	

Léon Walras: General Equilibrium Theory

Leon WALRAS (1834-1910)



(Wikipedia)



ÉLÉMENTS

D'ÉCONOMIE POLITIQUE

PURE

OU

THÉORIE DE LA RICHESSE SOCIALE

PAR

LÉON WALRAS

Professeur d'Économie politique à l'Académie de Lausanne.

LAUSANNE

IMPRIMERIE L. CORBAZ & Cie, ÉDITEURS

PARIS
GUILLAUMIN & Cie, ÉDITEURS
14, rue Richelieu.

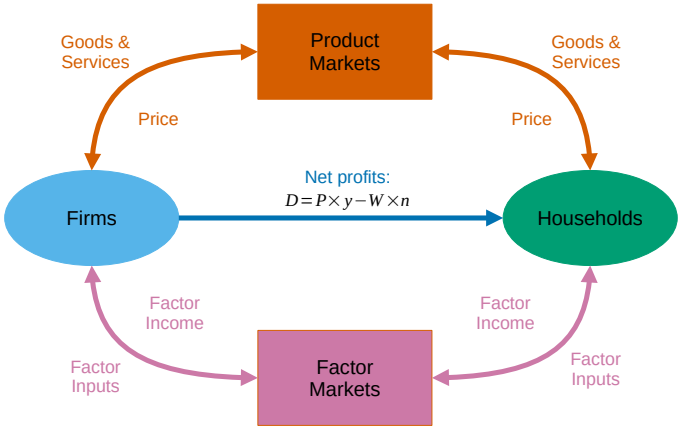
BALE
H. GEORG, LIBRAIRE-ÉDITEUR
Même maison à Genève.

1874

Tous droits réservés.

(1874)

Walrasian Equilibrium: Circular Flow

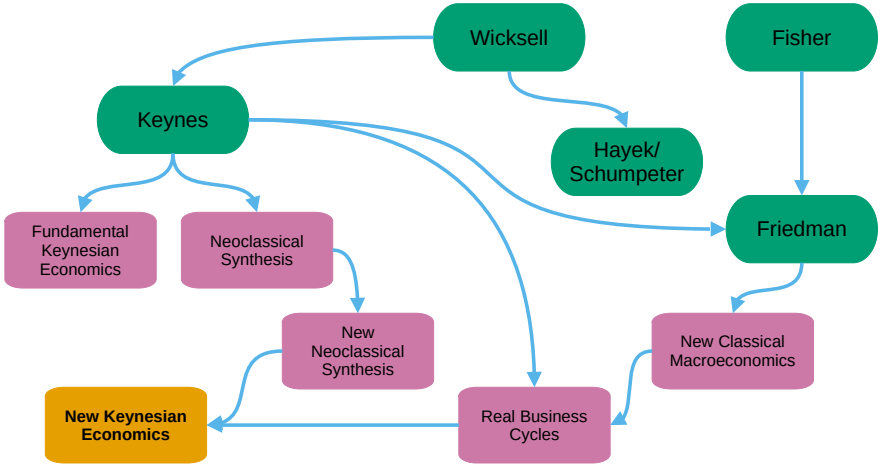


The Pre-KEYNESian Period

- Flexible prices
- Classical dichotomy of real and monetary sector: Monetary variables have no impact on real outcomes
- Classical quantity theory: $\overline{V} \times M = \overline{y} \times P$
- Monetary policy as technical issue until 1914 (gold standard)
- Business cycle theory only as informal reasoning

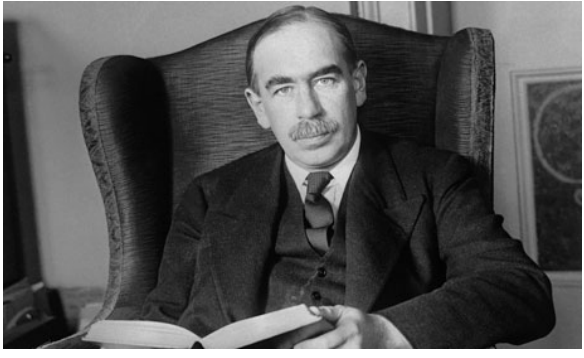
The Genealogy of Macroeconomics in the 20th Century

(Spahn 2016)

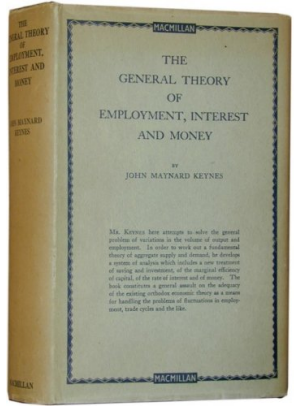


KEYNES' General Theory as a Milestone

John Maynard Keynes (1883-1946)



General Theory (1936)



The KEYNESian Revolution

- Classical economics cannot explain persistent unemployment during the Great Depression of the 1930s
- John Maynard KEYNES (1883-1946) as founder of macroeconomics: General Theory of Employment, Interest and Money (1936)
- General equilibrium analysis: simultaneous analysis of the key variables employment, income, interest rate and prices
- Insufficient aggregate demand can lead to persistent involuntary unemployment
- John HICKS (1904-1989): IS-LM model (1937)
- Large macroeconomic models for demand management

Can Pessimism Cause a Recession? Two Views

Households reduce consumption and increase desired saving.

What happens to GDP and employment?

Classical Benchmark

- Flexible prices and wages facilitate market clearing
- The real interest rate adjusts, stimulating investment
- Output and employment are determined by supply-side fundamentals

Keynesian View

- Prices and wages do not adjust sufficiently quickly
- Lower consumption reduces aggregate demand
- Firms reduce production and employment: a recession may result

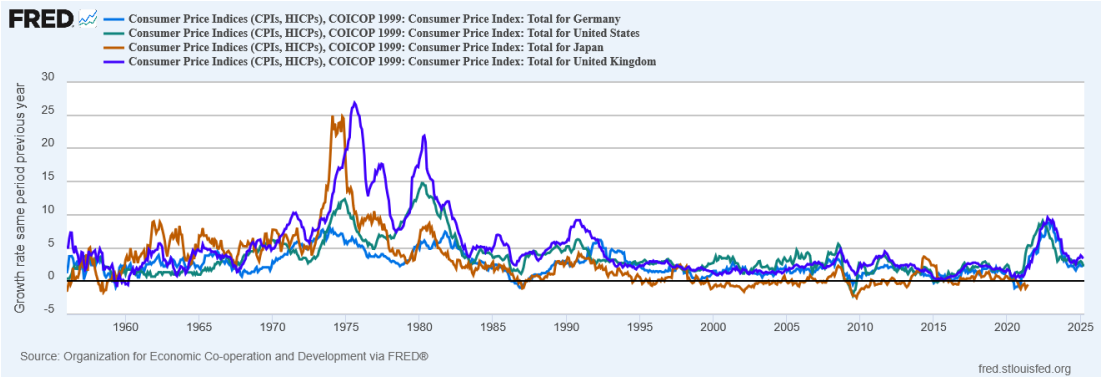
Different assumptions about economic adjustment lead to different predictions.

The Neoclassical Synthesis

- John R. HICKS (1904-1989), Paul A. SAMUELSON (1915-2009)
- IS-LM-Modell for short-run analysis with given prices and wages
- Classical dichotomy and quantity theory for the long-run (flexible prices)
- Don Patinkin (1922-1995): Adjustment from short to long run (Real balance effect)
- Phillips-curve: Apparent short-run trade-off between unemployment and inflation

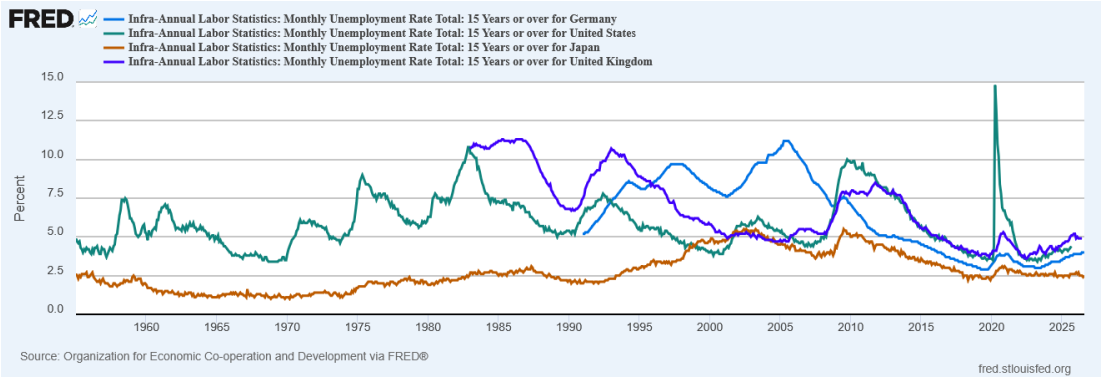
High Inflation in the 1970s

(FRED: CPALTT01∗M659N)



... and High Unemployment Rates

(FRED: LRHUTTTT∗M156S)



Stagflation: A Puzzle for the Phillips Curve

Conventional Phillips-curve reasoning:

Higher inflation is associated with lower unemployment

What happened in the 1970s?

At times, inflation and unemployment increased simultaneously

$$\pi \uparrow \qquad u \uparrow$$

- Does this observation invalidate the Phillips curve?
- What could change the relationship between inflation and unemployment?

Three Counter-Revolutions

- Three Counter-Revolutions since 1960:
  - Monetarism
  - New classical macroeconomics
  - Real business cycle theory

Monetarism

- Monetary disturbances are an important source of business cycle fluctuations (Brunner 1968)
- Monetary expansion can stimulate output and employment in the short run, but cannot permanently reduce unemployment below its natural rate
- Inflation expectations matter: there is no permanent trade-off between inflation and unemployment (Friedman 1968)
- Monetary policy operates with long and variable lags; Friedman advocated a constant money growth rule rather than discretionary fine-tuning

Explaining Stagflation: Expectations and Supply Shocks

How can inflation and unemployment rise simultaneously?

Expectations-Augmented Phillips Curve

$$\pi_t = \pi_t^e - \alpha(u_t - u_t^n) + s_t, \qquad \alpha > 0$$

π_t^e: expected inflation; u_t^n: natural rate of unemployment; s_t: cost-push shock

- Rising inflation expectations: Higher expected inflation shifts the short-run Phillips curve upward
- Adverse supply shocks: Higher production costs increase inflation and can raise unemployment simultaneously

Stagflation reflects a shift of the short-run Phillips curve, not necessarily a movement along it

New Classical Macroeconomics

- Rational expectations: Economic agents use available information and knowledge of the economic environment to form model-consistent expectations
- Anticipated vs. unanticipated policy: The effects of economic policy depend on whether policy actions are expected or come as a surprise
- Policy ineffectiveness proposition: In early New Classical models with market clearing and specific information assumptions, anticipated monetary policy does not affect real output
- Microfoundations: Aggregate economic relationships should be derived from optimizing individual behavior and equilibrium conditions

Robert E. Lucas, Thomas J. Sargent, Neil Wallace

The Lucas Critique: Can We Trust Historical Relationships?

An Empirical Relationship

An economist estimates a Phillips curve using historical data:

$$\pi_t = a - bu_t + \varepsilon_t, \qquad b > 0$$

A Change in the Policy Regime

The central bank announces a permanent increase in its inflation target

- Can we use the estimated coefficients a and b to predict the effects of the new policy?
- How might households' and firms' expectations and decisions change?

Empirical macroeconomic relationships may change when the policy regime changes

Real Business Cycle Theory: Macroeconomic Fluctuations as Equilibrium Outcomes

- Microfounded dynamic equilibrium: Households and firms make optimal intertemporal decisions under uncertainty
- Real shocks: Business cycle fluctuations are driven primarily by productivity shocks and other real disturbances
- Flexible prices and market clearing: Prices adjust to clear markets; Money is neutral in the benchmark model
- Efficient fluctuations: In the frictionless competitive benchmark, business cycles are efficient equilibrium responses to economic shocks

Kydland and Prescott (1982)

Can Monetary Policy Stabilize an RBC Economy?

Suppose an adverse productivity shock reduces aggregate output

Can expansionary monetary policy offset the real effects of this shock in the benchmark RBC model?

Explain your reasoning.

2. History of Macroeconomics

Traditional Macroeconomics
Modern Macroeconomics
Criticisms of Macroeconomics
Summary and Literature

Market Imperfections and Macroeconomic Fluctuations

- Nominal rigidities: Prices and wages do not adjust immediately to changing economic conditions
- Imperfect competition: Firms have market power and may set prices above marginal costs
- Monetary non-neutrality: With nominal rigidities, monetary policy can affect real output and employment in the short run
- Inefficient fluctuations: Market imperfections can make equilibrium allocations inefficient, creating a potential role for welfare-improving stabilization policy

The New Neoclassical Synthesis

Common Methodological Foundations

- Intertemporal optimization, rational expectations, and dynamic general equilibrium

	RBC	New Keynesian
Nominal rigidities	No	Yes
Monetary non-neutrality	No	Short run
Inefficient fluctuations	No	Possible

The New Neoclassical Synthesis combines the dynamic general equilibrium framework with New Keynesian market imperfections

What Do We Need to Build a Macroeconomic Model?

The Policy Question

Suppose the central bank unexpectedly raises its policy interest rate

We want to predict the effects on GDP, inflation, and employment

What ingredients should our model contain?

- Economic agents: Whose decisions must we model, and what determines their behavior?
- Markets and equilibrium: How do individual decisions interact, and what conditions must hold in equilibrium?
- Dynamics and shocks: Why do expectations about the future matter, and how do we represent unexpected disturbances?

Dynamic Stochastic General Equilibrium (DSGE) Models

- Dynamic: Current decisions depend on past conditions and expectations about the future
- Stochastic: Economic activity is affected by random shocks and uncertainty
- General Equilibrium: Decisions of households, firms, and other agents are mutually consistent, and markets clear; equilibrium does not imply efficiency or full employment
- Microfoundations: Behavioral equations are derived from individual objectives and constraints

Key Insight

DSGE is a modeling framework, not a single theory of the economy; different assumptions lead to different predictions

2. History of Macroeconomics

Traditional Macroeconomics
Modern Macroeconomics
Criticisms of Macroeconomics
Summary and Literature

The Trouble with Macroeconomics

(Paul Romer 2016)

"Lee Smolin begins The Trouble with Physics (Smolin 2007) by noting that his career spanned the only quarter-century in the history of physics when the field made no progress on its core problems. The trouble with macroeconomics is worse. I have observed more than three decades of intellectual regress."

Where Modern Macroeconomics Went Wrong

(Stiglitz 2018)

"Dynamic Stochastic General Equilibrium (DSGE) models, which have played such an important role in modern discussions of macroeconomics, in my judgement fail to serve the functions which a well-designed macroeconomic model should perform."

Are DSGE Models Useful?

- Restrictive assumptions: Representative agents, rational expectations, and simplified market structures may neglect important economic mechanisms
- Missing mechanisms: Pre-crisis benchmark models often neglected financial frictions, instability, and heterogeneous agents
- Empirical credibility: Reliance on unobserved shocks and difficult-to-test assumptions raises questions about identification and validation
- Policy relevance: Models that omit important mechanisms may deliver misleading predictions and policy recommendations

Key Distinction

Criticism of particular DSGE models does not necessarily invalidate the DSGE modeling framework

After the Storm: How DSGE Models Have Evolved

(Christiano, Eichenbaum and Trabandt 2018)

- Financial frictions: Credit constraints, financial intermediaries, and balance-sheet effects can amplify economic shocks
- Nonlinearities: Occasionally binding constraints and the effective lower bound on interest rates can change economic dynamics
- Heterogeneous agents: Differences in income, wealth, and financial constraints affect the transmission of shocks and policies
- Empirical discipline: Model predictions must be confronted with data and evaluated against alternative explanations

Key Insight

The DSGE framework continues to evolve, but greater model complexity does not automatically imply better models

2. History of Macroeconomics

Traditional Macroeconomics
Modern Macroeconomics
Criticisms of Macroeconomics
Summary and Literature

From the History of Macroeconomics to Modern Modeling

- Economic puzzles drive theoretical progress: Macroeconomic models evolve in response to empirical evidence and limitations of existing theories
- Methodological convergence: Modern macroeconomics combines intertemporal optimization, expectations, stochastic shocks, and general equilibrium analysis
- Assumptions matter: Model predictions and policy recommendations depend on the economic mechanisms included and their empirical relevance

What Comes Next?

We will develop the building blocks of dynamic macroeconomic models:

- Static general equilibrium (Chapter 3)
- Intertemporal decisions (Chapter 4)

Literature

History of Macroeconomics


Blanchard, O. (2017): Macroeconomics, 7th Edition, Pearson, Chapter 24


Burda, M.; Wyplosz, Ch. (2022): Macroeconomics. A European Text, 8th Edition, Oxford University Press, Chapter 20


Heilbroner, R.L. (2000): The Worldly Philosophers. The Lives, Times and Ideas of the Great Economic Thinkers, Penguin Books


Spahn, P. (2016): Streit um die Makroökonomie. Theoriegeschichtliche Debatten von Wicksell bis Woodford, Metropolis-Verlag

Literature

Criticism and recent developments

- Akerlof, G. A. (2002): Behavioral macroeconomics and macroeconomic behavior, American Economic Review 92(3), 411-433
- Blanchard, O. (2009): The state of macro, Annual Review of Economics 1(1), 209-228
- Caballero, R. J. (2010): Macroeconomics after the crisis. How to deal with the pretense-of-knowledge syndrome, NBER Working Paper 16429
- Christiano, L.J.; Eichenbaum, M.S.; Trabandt, M. (2018): On DSGE models, *Journal of Economic Perspectives* 32(3), 113-140
- Romer, P. (2016): The trouble with macroeconomics, https://paulromer.net/the-trouble-with-macro/WP-Trouble.pdf
- Stiglitz, J.E. (2018): Where modern macroeconomics went wrong, *Oxford Review of Economic Policy* 34(1-2), 70-106